

Data Analysis

Task 3: Classification

Tomas Raila

Vilnius University
Institute of Computer Science

March 8, 2024

Task description

- ▶ Dataset
 - ▶ Pick the dataset you have previously selected for the 1st task.
 - ▶ If, for some reason, the dataset is unsuitable for this task - you can select another (but confirm it with Tomas)
- ▶ DIY classification:
 - ▶ Implement a classification algorithm of your choice **from scratch**.
 - ▶ Popular choices: logistic regression, decision tree, k-nearest neighbors...
 - ▶ Compare your implementation with **scikit-learn** version and show that it works correctly.

Task description (2)

- ▶ Feature selection/engineering
 - ▶ Choose one binary categorical variable (or more) which you want to predict using classification models.
 - ▶ You can define a custom variable by using other continuous/categorical variables.
 - ▶ Perform any data preprocessing if necessary - remove outliers, fix missing data, etc...
 - ▶ Perform feature selection/engineering: various feature combinations, polynomial features, one-hot encoding, etc...
 - ▶ See if any findings from 1st task (EDA) could be applied to help select suitable features.
 - ▶ Check class sizes and balance them if necessary - apply strategies of your choice.

Task description (3)

- ▶ Model evaluation & tuning
 - ▶ Choose at least 2 classification algorithms (from scikit-learn).
 - ▶ Train at least 2 classification models using each algorithm and different feature sets (4 different models in total).
 - ▶ Evaluate and compare model performance: use metrics of your choice, confusion matrices, etc...
 - ▶ Choose **one** model and try to tune it for better performance on your classification problem: adjust hyperparameters, apply regularization, change classification threshold, etc...
- ▶ Result interpretation
 - ▶ 5 most important conclusions in domain terms
 - ▶ Connection between the numbers and charts and your own knowledge
 - ▶ Any unexpected, surprising or otherwise interesting facts about the data? Can you prove/disprove any of your initial assumptions based on the results?

Other info

- ▶ Maximum grade for the task: **1.0**
- ▶ Deadline for task submission to Gitlab: **2024-03-12 16:00**
- ▶ Task submission process:
 - ▶ Jupyter notebook + dataset (.csv file) uploaded to Gitlab, according to the instructions presented in lecture 1
 - ▶ Task presented and explained during exercise class (Tuesdays/Fridays, 18:00 - 19:30)